

## Introduction to Quantum Computing

### Key Images: Week 1

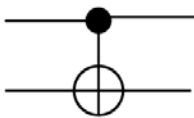
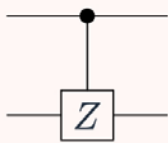
#### Quantum Gates > List of Classical Gates

| GATE |                                                                                           | CIRCUIT REPRESENTATION                                                              | TRUTH TABLE                       |                            |
|------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------|----------------------------|
| NOT  | The output is 1 when the input is 0 and 0 when the input is 1.                            |    | Input<br>0<br>1                   | Output<br>1<br>0           |
| AND  | The output is 1 only when both inputs are 1, otherwise the output is 0.                   |    | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>0<br>0<br>0<br>1 |
| OR   | The output is 0 only when both inputs are 0, otherwise the output is 1.                   |  | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>0<br>1<br>1<br>1 |
| NAND | The output is 0 only when both inputs are 1, otherwise the output is 1.                   |  | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>1<br>1<br>1<br>0 |
| NOR  | The output is 1 only when both inputs are 0, otherwise the output is 0.                   |  | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>1<br>0<br>0<br>0 |
| XOR  | The output is 1 only when the two inputs have different value, otherwise the output is 0. |  | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>0<br>1<br>1<br>0 |
| XNOR | The output is 1 only when the two inputs have the same value, otherwise the output is 0.  |  | Input<br>0 0<br>0 1<br>1 0<br>1 1 | Output<br>1<br>0<br>0<br>1 |

## Quantum Gates > Single Qubit Gates

| GATE                                                                                                                                                                                                         | CIRCUIT REPRESENTATION                   | MATRIX REPRESENTATION                                                  | TRUTH TABLE                                                                                                                                                                                                                                                   | BLOCH SPHERE |        |             |                                          |             |                                          |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------|-------------|------------------------------------------|-------------|------------------------------------------|--|
| <i>I</i> Identity-gate:<br>no rotation is performed.                                                                                                                                                         |                                          | $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$                     | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math> 0\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math> 1\rangle</math></td></tr></table>                                                           | Input        | Output | $ 0\rangle$ | $ 0\rangle$                              | $ 1\rangle$ | $ 1\rangle$                              |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $ 0\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $ 1\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>X</i> gate:<br>rotates the qubit state by $\pi$ radians ( $180^\circ$ ) about the x-axis.                                                                                                                 |                                          | $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$                     | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math> 1\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math> 0\rangle</math></td></tr></table>                                                           | Input        | Output | $ 0\rangle$ | $ 1\rangle$                              | $ 1\rangle$ | $ 0\rangle$                              |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $ 1\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $ 0\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>Y</i> gate:<br>rotates the qubit state by $\pi$ radians ( $180^\circ$ ) about the y-axis.                                                                                                                 |                                          | $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$                    | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math>i 1\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math>-i 0\rangle</math></td></tr></table>                                                        | Input        | Output | $ 0\rangle$ | $i 1\rangle$                             | $ 1\rangle$ | $-i 0\rangle$                            |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $i 1\rangle$                             |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $-i 0\rangle$                            |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>Z</i> gate:<br>rotates the qubit state by $\pi$ radians ( $180^\circ$ ) about the z-axis.                                                                                                                 |                                          | $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$                    | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math> 0\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math>- 1\rangle</math></td></tr></table>                                                          | Input        | Output | $ 0\rangle$ | $ 0\rangle$                              | $ 1\rangle$ | $- 1\rangle$                             |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $ 0\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $- 1\rangle$                             |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>S</i> gate:<br>rotates the qubit state by $\frac{\pi}{2}$ radians ( $90^\circ$ ) about the z-axis.                                                                                                        |                                          | $S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$    | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math> 0\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math>e^{i\frac{\pi}{2}} 1\rangle</math></td></tr></table>                                         | Input        | Output | $ 0\rangle$ | $ 0\rangle$                              | $ 1\rangle$ | $e^{i\frac{\pi}{2}} 1\rangle$            |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $ 0\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $e^{i\frac{\pi}{2}} 1\rangle$            |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>T</i> gate:<br>rotates the qubit state by $\frac{\pi}{4}$ radians ( $45^\circ$ ) about the z-axis.                                                                                                        |                                          | $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$    | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math> 0\rangle</math></td></tr><tr><td><math> 1\rangle</math></td><td><math>e^{i\frac{\pi}{4}} 1\rangle</math></td></tr></table>                                         | Input        | Output | $ 0\rangle$ | $ 0\rangle$                              | $ 1\rangle$ | $e^{i\frac{\pi}{4}} 1\rangle$            |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $ 0\rangle$                              |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $e^{i\frac{\pi}{4}} 1\rangle$            |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| <i>H</i> gate:<br>rotates the qubit state by $\pi$ radians ( $180^\circ$ ) about an axis diagonal in the x-z plane. This is equivalent to an X-gate followed by a $\frac{\pi}{2}$ rotation about the y-axis. |                                          | $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 0\rangle</math></td><td><math>\frac{ 0\rangle +  1\rangle}{\sqrt{2}}</math></td></tr><tr><td><math> 1\rangle</math></td><td><math>\frac{ 0\rangle -  1\rangle}{\sqrt{2}}</math></td></tr></table> | Input        | Output | $ 0\rangle$ | $\frac{ 0\rangle +  1\rangle}{\sqrt{2}}$ | $ 1\rangle$ | $\frac{ 0\rangle -  1\rangle}{\sqrt{2}}$ |  |
| Input                                                                                                                                                                                                        | Output                                   |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 0\rangle$                                                                                                                                                                                                  | $\frac{ 0\rangle +  1\rangle}{\sqrt{2}}$ |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |
| $ 1\rangle$                                                                                                                                                                                                  | $\frac{ 0\rangle -  1\rangle}{\sqrt{2}}$ |                                                                        |                                                                                                                                                                                                                                                               |              |        |             |                                          |             |                                          |  |

## Quantum Gates > Two-Qubit Gates

| GATE                                                                                                      | CIRCUIT REPRESENTATION                                                            | MATRIX REPRESENTATION                                                                                   | TRUTH TABLE                                                                                                                                                                                                                                                                                                                                                |       |        |              |              |              |              |              |              |              |               |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|
| Controlled-NOT gate:<br>apply an X-gate to the target qubit if the control qubit is in state $ 1\rangle$  |  | $CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 00\rangle</math></td><td><math> 00\rangle</math></td></tr><tr><td><math> 01\rangle</math></td><td><math> 01\rangle</math></td></tr><tr><td><math> 10\rangle</math></td><td><math> 11\rangle</math></td></tr><tr><td><math> 11\rangle</math></td><td><math> 10\rangle</math></td></tr></table>  | Input | Output | $ 00\rangle$ | $ 00\rangle$ | $ 01\rangle$ | $ 01\rangle$ | $ 10\rangle$ | $ 11\rangle$ | $ 11\rangle$ | $ 10\rangle$  |
| Input                                                                                                     | Output                                                                            |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 00\rangle$                                                                                              | $ 00\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 01\rangle$                                                                                              | $ 01\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 10\rangle$                                                                                              | $ 11\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 11\rangle$                                                                                              | $ 10\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| Controlled-phase gate:<br>apply a Z-gate to the target qubit if the control qubit is in state $ 1\rangle$ |  | $cZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$  | <table><tr><th>Input</th><th>Output</th></tr><tr><td><math> 00\rangle</math></td><td><math> 00\rangle</math></td></tr><tr><td><math> 01\rangle</math></td><td><math> 01\rangle</math></td></tr><tr><td><math> 10\rangle</math></td><td><math> 10\rangle</math></td></tr><tr><td><math> 11\rangle</math></td><td><math>- 11\rangle</math></td></tr></table> | Input | Output | $ 00\rangle$ | $ 00\rangle$ | $ 01\rangle$ | $ 01\rangle$ | $ 10\rangle$ | $ 10\rangle$ | $ 11\rangle$ | $- 11\rangle$ |
| Input                                                                                                     | Output                                                                            |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 00\rangle$                                                                                              | $ 00\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 01\rangle$                                                                                              | $ 01\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 10\rangle$                                                                                              | $ 10\rangle$                                                                      |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |
| $ 11\rangle$                                                                                              | $- 11\rangle$                                                                     |                                                                                                         |                                                                                                                                                                                                                                                                                                                                                            |       |        |              |              |              |              |              |              |              |               |